

RoboTrader

Remediation and Launch Plan

Safety-first path from contained paper/read-only operation to an explicitly gated limited live canary

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Status: Active execution plan; Gate A remains closed

Source baseline: repository audit at commit `51f0e99` on branch `main`

Target: safe supervised paper operation first, then remote read-only access, then an explicitly gated limited live canary

Operational status: TRADER REMAINS STOPPED UNTIL CUMULATIVE GATE A PASSES

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Current execution baseline (2026-07-25): `main` includes the truthful Phase 0 CI gates from PR #83 (7f5de0a) and runtime-stability fixes from PR #81 (b7e5005). PR #82 completed PR 1 and merged as 393f533. PR #87 completed PR 1A and merged as 4caf782cbf43ff4397f1b89b42d5f657ecee8e, closing the incident-driven broker-correlation and protected-runtime gate. PR #80 is superseded and must not be merged or cherry-picked; its independent findings are assigned to later scoped PRs in `docs/branch_analysis/PR80_DISPOSITION_2026-07-23.md`. PR #90 completed the PR 1B diagnostic implementation and merged as 0d43006561071e27b217cb6d16f3c0a245a18655. Its first real invocation from merged `main` failed closed before broker connection because the local runtime does not yet define an explicit paper-account identity and allow-list. Issue #92 tracks that local-only operator prerequisite. Protected evidence hashes were unchanged and no correction was attempted. PR #95 then completed the PR 1C package/import prerequisite and merged as dff4c8b597a54c614d8925565f28aa865f8ae676. The package root is now inert, active subpackages and the required dashboard template are included in built wheels, broker safeguards remain explicit at direct IB users, and runtime artifacts and secrets remain excluded. Runtime dependency metadata and a true clean-install wheel gate remain assigned to PR 9. Gate A remains closed, the trader remains stopped. Dormant safety-core PR 2A merged through PR #97 as d17b0d5b4f31ab15e2a9b138cca006c0103b7276; it has no production imports or runtime wiring. PR 2B.1 merged through PR #104 as 3ecd05b3352ddcd4519662b0fe957751f3fdb1 from exact reviewed head 0d5585b46f8f1b495d944e24b23df5a7c01cfc2d. It binds the dormant paper safety coordinator to startup replay but deliberately leaves production order authorization and submission unwired. PR 2B.2 is next. Gate A remains closed and this merge is not a launch authorization.

1. Purpose

This document turns the full codebase audit into an ordered implementation program. It is intended to remain the reference plan for future pull requests, reviews, test campaigns, launch decisions, and operational sign-off.

The ordering is deliberate. Later work must not begin merely because earlier code was merged. Every phase has evidence gates that must pass in the actual supported runtime.

The plan addresses:

- trading correctness;
- IBKR and exchange integration;
- order execution and paper/live separation;
- risk, sizing, stops, take-profit, and strategies;
- backtesting and market-data quality;
- database integrity, backups, and reconciliation;
- security, secrets, identity, authorization, and API safety;
- logging, auditability, testing, CI/CD, packaging, and deployment;
- dashboard, mobile, cloud, missing features, and unfinished features.

2. Non-negotiable rules

1. Keep `IBKR ReadOnlyApi=yes` until the live-enablement gate in PR 11 has been implemented and independently approved.
2. Never delete or rewrite user trading data without explicit user authorization, a verified backup, a preview of the exact impact, and a tested rollback path.
3. Do not treat a green unit-test count as launch evidence. Launch evidence includes reconciliation, restart, restore, failure-injection, alert, and soak results.
4. Broker state becomes authoritative whenever real broker orders are possible. Local database state alone must never authorize trading after startup or reconnect.
5. A kill switch must block risk-increasing orders while preserving a narrowly defined reduce-only path.
6. No remote control, mobile control, or cloud exposure is allowed before identity, authorization, TLS, WebSocket privilege separation, and durable audit controls exist.
7. Backtests cannot approve a strategy until the accounting and execution model in PR 6 is complete.
8. Every production claim must name the supported topology. Until changed by an approved architecture decision, that topology is local macOS, IBC, IB Gateway paper account, and the authoritative launcher.
9. Every PR must be independently reviewable, revertible where practical, and leave the paper system in a coherent state.
10. Incomplete strategies and alternate engines remain disabled or quarantined rather than partially integrated.

3. Current baseline and containment boundary

The current sanctioned runtime is materially contained:

- `START_TRADER.sh` uses paper port 4002 and validates IBC read-only configuration.
- `runner_async.py` constructs `PaperExecutor`.
- The active subprocess broker client is read-oriented and does not expose a complete order-placement interface.
- The current local dashboard is bound to loopback.
- The preflight gate, launchd watchdog, persistent connection recovery, test isolation, and persisted kill switch are valuable foundations.

This containment must be made explicit and tested in PR 1. It must not be weakened while the later live-order architecture is built.

4. How to use this plan

For each PR:

1. Create one tracking issue using the PR section below as its body.
2. Confirm its prerequisites are complete and evidenced.
3. Add or update tests before changing safety behavior where feasible.
4. Implement only the stated scope. Record extra discoveries in the problem register rather than silently expanding the PR.
5. Run the PR-specific verification suite plus the repository regression suite.
6. Attach evidence to the PR: commands, test output, screenshots where applicable, migration/restore results, and fault-injection results.
7. Update this document's progress register after merge.
8. Do not advance a launch gate until every required PR and operational exercise for that gate is complete.

Recommended branch naming: `codex/pr-XX-short-name` or the team's equivalent feature prefix.

5. Dependency sequence

The required order is:

1. PR 1 preserves the safe paper-only boundary and removes destructive hazards.
2. PR 1A closes the active cross-symbol market-data contamination path before any safety exit can be allowed through a kill switch.
3. PR 1B adds non-mutating broker-versus-ledger reconciliation and resolves the incident evidence gate without clearing safety state.
4. PR 1C makes package-root imports side-effect-free and ensures the dormant safety core and all active subpackages can ship in built wheels.
5. PRs 2 through 5 make active paper behavior safe and state authoritative.
6. PR 6 repairs the evidence engine used to evaluate strategies.
7. PR 7 unifies strategy and risk behavior on top of corrected data and state.
8. PRs 8 through 10 establish remote security, reproducible delivery, and honest operations.
9. PRs 11 and 12 add live order placement and broker-native protection behind a disabled gate.
10. PR 13 proves failure behavior through soak and fault injection.
11. PR 14 delivers an explicitly supported remote/cloud topology if still required.
12. PR 15 permits only a tiny, manually armed live canary.

PRs may be prepared in parallel only when their files and safety contracts do not overlap. Merge order still follows the dependency chain.

Phase A - Preserve containment and correct the active paper system

PR 1 - Freeze live mode and quarantine destructive utilities

Objective

Make the current paper/read-only boundary unambiguous, machine-enforced, and visible. Remove the possibility that a dormant script can destroy user data during later remediation work.

Problems addressed

- Conflicting `EXECUTION_MODE`, `TRADING_MODE`, `TRADING_ENV`, and deployment-mode variables.
- Production manifests that claim live operation while the runner remains paper.
- Missing IBC-config branch that can allow startup checks to pass when the file is absent.
- Dashboard status that can report a hardcoded or incorrect mode.
- Destructive recovery and IB position-sync scripts.
- Shared paper/live database, logs, artifacts, or credentials.

Step-by-step work

1. Define one canonical runtime contract containing environment, execution mode, broker port, broker account identifier, broker account type, database path, model-artifact set, and build identifier.
2. Add a startup validator that rejects every inconsistent combination, including:
 - paper mode with live port 4001;
 - live mode with paper port 4002;
 - missing IBC configuration;
 - IBC `ReadOnlyApi` not equal to `yes` during the containment phase;
 - missing or unapproved broker account identifier;
 - shared paper/live database path;
 - production mode without authentication, signing, alerting, or backup readiness.
3. Make the authoritative launcher and runner refuse live execution regardless of environment values. Use a separate compile-time or capability flag that remains disabled until PR 11.
4. Replace dashboard hardcoded mode reporting with the validated server-side runtime contract.
5. Add an unmistakable PAPER banner containing account alias, execution source, database identity, build SHA, and configuration fingerprint.
6. Inventory every alternate launcher. Disable or route `scripts/start_runner.sh`, Docker commands, dashboard start, and Kubernetes commands through the same contract.
7. Quarantine `simple_recover.py`, `recover_database.py`, and `sync_ib_positions.py` so they cannot run accidentally.
8. Design replacement maintenance commands with:
 - read-only preview;
 - portfolio-scoped diff;
 - SQLite online backup;
 - explicit typed confirmation;
 - transaction boundaries;
 - post-operation integrity and reconciliation checks.
9. Document the current supported topology and explicitly label Docker/Kubernetes/live as unsupported.

Required tests

- Table-driven tests for every mode/port/account/database combination.
- Test that missing IBC configuration fails before connecting.
- Test that all sanctioned endpoints invoke the same validation.
- Test that live mode remains impossible even with contradictory environment variables.
- Test that maintenance utilities default to preview and cannot delete without the authorization sequence.

Done means

- No sanctioned command can connect to port 4001 or create a live executor.
- Dashboard and logs derive mode from the validated runtime contract.
- Paper and potential live state paths cannot collide.
- Destructive utilities cannot modify data by import or by default invocation.
- The supported-topology document matches executable behavior.

Rollback

Revert only to the previous paper launcher. Never restore destructive utility behavior.

PR 1A - Fail closed on broker-data correlation

Objective

Prevent a delayed or timed-out IBKR response from being assigned to another symbol, and prevent malformed historical timestamps from overwriting the evidence needed to detect contamination.

This is an incident-driven prerequisite discovered on 2026-07-23. A timed-out historical-data request left a response in an uncorrelated FIFO queue. Later requests consumed shifted responses, a quote for one symbol was labeled as another symbol, and the mislabeled value immediately triggered position-loss and stop logic. The stop did not execute because the kill switch blocked all orders. Therefore PR 2 must not enable reduce-only exits until PR 1A is merged and verified.

Problems addressed

- RT-010 and RT-033.
- Subprocess requests and responses have no correlation identifier.
- A timeout does not invalidate or drain the uncertain worker session.
- Market-data responses are trusted without contract-symbol or could verification.
- Historical bars persist a DataFrame RangeIndex instead of normalized market timestamps, so later cycles overwrite history.
- Cross-symbol contamination can reach risk checks, stop-loss logic, account values, and the database.
- Historical bars are not a live protective feed, and no independent

protective-price writer exists before PR 3. Existing holdings must therefore abort startup instead of running with empty or stale stop-monitor prices.

- Position-load and stop-registration failures can currently be logged and treated as empty or partially protected state.
- A fail-closed setup abort can leak resources or be restarted repeatedly by the watchdog unless cleanup and supervision preserve the safety decision.

Step-by-step work

1. Version every command and response envelope with a worker-generation

identifier, request identifier, and command name.

2. Match responses by request and worker-generation identifier; reject missing, duplicate, stale, malformed, or unexpected identifiers.

3. After any timeout or protocol uncertainty, terminate and recreate the worker session and all response state before accepting another response. Never automatically retry the uncertain request.

4. Include requested and returned contract identity in each response: symbol, conId, exchange, currency, request type, and observation timestamp.

5. Reject mismatched or incomplete identities before updating risk, stops, positions, account values, caches, or persistent state.

6. Normalize historical bars to validated timezone-aware timestamps from the broker payload. Reject RangeIndex, duplicate, non-monotonic, future, or session-invalid timestamps.

7. Make persistence append or upsert by the validated timestamp and contract identity; never overwrite another symbol or cycle silently.

8. Pass broker event time into freshness checks. Receipt time must not make a stale bar appear current.

9. Poison and disconnect the client generation on timeout, cancellation, malformed envelopes, reader failure, or identity mismatch. Reject queued and new commands until recovery creates an isolated generation.

10. Abort the remaining symbol cycle after protocol poison and surface the failure to connection health.

11. Until PR 3 supplies and confirms an independent live protective feed, fail setup for every nonzero existing position unless the stop monitor owns a matching pending stop and a fresh accepted `live_protective` event.

12. Treat database position-load and stop-registration uncertainty as fatal; never fall back to an empty holdings view.

13. Clean every partially initialized resource independently, exit nonzero with a sanitized audit reason, and make the watchdog suppress automatic restart for that exact terminal safety exit until an operator deliberately runs the authoritative launcher after protection is restored.

Required tests

- A delayed response after timeout cannot satisfy the next request.
- Out-of-order, duplicate, missing-ID, and unknown-ID responses fail closed.
- A symbol or conId mismatch cannot reach risk or stop callbacks.
- Timeout recovery starts a clean worker session and cannot reuse queued data.
- Cancellation followed by a late response cannot affect another request.
- Missing, wrong, duplicate, stale-generation, or malformed envelopes poison the transport.
- A deterministic three-symbol sequence proves no response shifting.
- RangeIndex and malformed timestamps are rejected.
- Valid timezone-aware bars persist without cross-cycle overwrite.
- Cross-symbol payload duplication is detected before persistence.
- Protocol failure produces no database write, cache update, latest-price update, strategy call, trailing-stop adjustment, or stop execution.

- Existing holdings with no monitor-owned live event, a historical-only event, a missing or mismatched stop, or stale/future timestamps fail setup.
- Position-load and stop-registration failures are fatal.
- A stop-monitor cleanup failure cannot prevent IBKR and database cleanup.
- The watchdog policy suppresses the terminal unprotected-position restart and permits ordinary nonterminal recovery.

Done means

- Every accepted broker-data response is bound to the exact originating request and contract.
- A timeout cannot contaminate a later request.
- Invalid data cannot reach trading, risk, stop, valuation, or persistence consumers.
- Historical rows retain auditable market timestamps.
- The incident scenario passes deterministic failure-injection tests.
- A holdings-bearing runner cannot look healthy while its stops are blind, and its terminal safety abort survives the process-supervisor boundary.

Non-goals

- Do not migrate or rewrite historical rows.
- Do not correct positions, account values, kill-switch state, or lock files.
- Do not connect to the broker while running unit and failure-injection tests.
- Defer broader source lineage, session policy, and real-time feed unification to PR 3.

PR 1B - Add read-only broker-ledger reconciliation

Objective

Produce the broker evidence required to explain the incident and determine whether a separate user-approved, backed-up state-correction action should be designed, without placing orders or mutating local data.

Problems addressed

- No safe read-only broker-versus-ledger reconciliation command exists.
- Preflight guidance points to a nonexistent reconciliation script.
- The old synchronization utility is intentionally inert because it previously deleted and replaced positions.
- Local paper-executor fills and the IBKR paper account may represent different systems, but the difference is not made explicit.

Step-by-step work

1. Create a diagnostic-only command that explicitly requests an IBKR read-only session on paper port 4002.
2. Reuse the PR 1 runtime contract and verify the managed account exactly before reading account data.
3. Read and mask account identity, contract identity, positions, average cost,

open orders, and recent executions.

4. Open SQLite in immutable or read-only mode and compute a portfolio-scoped diff without invoking application write paths.

5. Report source, timestamp, freshness, symbol, conId, exchange, currency, quantity, cost, orders, and executions for each difference.

6. Make order submission methods unavailable by construction and fail if the Gateway or client cannot prove read-only operation.

7. Prove that database, kill-switch, lock, bypass-log, and trading-log hashes remain unchanged after the command.

8. Replace the nonexistent preflight remediation reference with this command.

9. Document that output is evidence only: it cannot clear a kill switch, correct the ledger, or authorize startup.

10. Run the affected-symbol reconciliation and obtain explicit operator approval before designing any separate state-correction action.

Required tests

- Paper port and read-only flags are mandatory and cannot be overridden.
- The managed account must match the runtime contract.
- Account identifiers and sensitive fields are masked in normal and error

output.

- No order method is reachable.
- SQLite is opened read-only and every relevant file hash remains unchanged.
- Quantity, cost, contract, open-order, and recent-execution differences are deterministic.
- Missing, stale, or ambiguous broker data fails closed.
- Preflight points only to the implemented diagnostic command.

Done means

- A broker-versus-ledger snapshot can be produced without mutation.
- The incident evidence has an explicit broker/account interpretation.
- Any proposed state correction is a separate reviewed action requiring user

authorization and backup.

Operational gate

The trader remains stopped after PR 1A and PR 1B. Reconciliation and operator review are necessary evidence, but they do not authorize startup. The first supervised paper start requires cumulative Gate A: PRs 1, 1A, 1B, 2 through 5, the relevant PR 7 controls, reviewed reconciliation, restore evidence, and all ordinary `. /START_TRADER . sh` checks passing without deleting or bypassing safety state.

Current operational evidence (2026-07-23): the merged command failed closed before broker connection because the local runtime has no explicit paper account identity or allow-list. Issue #92 tracks the local-only configuration prerequisite. The blocked run changed none of the protected evidence files and does not satisfy the reviewed-reconciliation requirement above.

PR 1C - Isolate package imports and complete wheel discovery

Objective

Remove two structural blockers discovered during the adversarial design review for dormant safety-core issue #91: importing `robo_trader.safety` must not load or mutate broker code, and a built wheel must not omit the safety package or other active subpackages.

Problems addressed

- Python executes `robo_trader/__init__.py` before any nested package import, and the previous initializer eagerly imported `ibkr_safe`, loaded `ib_async`, and globally patched `IB.disconnect`.
- `pyproject.toml` declared only the root `robo_trader` package, so source-tree tests could pass while built wheels omitted every nested package.
- Broad package discovery could accidentally include archived code, similarly named sibling packages, tests, configuration, credentials, databases, or logs.
- Required non-Python assets could be omitted even when their Python package was included.

Step-by-step work

1. Make `robo_trader/__init__.py` metadata-only and side-effect-free.
2. Require every production module that directly constructs `ib_async.IB` to activate `ibkr_safe` explicitly.
3. Replace the root-only package list with precise, namespace-aware discovery for `robo_trader` and `robo_trader.*`.
4. Exclude archived code and similarly named sibling packages.
5. Disable implicit package-data inclusion and allow-list only the required bug-dashboard template.
6. Require a patched `setuptools` build backend and pin the supported development backend.
7. Add cold-import tests that reject broker imports, filesystem mutation, sockets, subprocesses, threads, current-directory changes, and environment changes.
8. Build a wheel from a copied exact tree, inspect its contents, render the packaged dashboard template, and import representative regular and namespace subpackages from outside the checkout.

Required tests

- A cold `python -I` root import is inert and does not load broker modules.
- Explicit `ibkr_safe` activation remains idempotent and force-only.
- Every direct production IB user activates the disconnect guard.
- Built wheels include active regular and namespace packages plus the required dashboard template.
- Built wheels exclude archives, sibling packages, tests, config, `.env`, IBC credentials, databases, and logs.
- The full repository suite and supported Python 3.10 through 3.12 hosted matrices pass.

Done means

- The package root is safe for PR 2A to add and import

`robo_trader.safety` without activating broker behavior.

- A built wheel contains the dormant safety core and all other active

subpackages once they are added.

- Direct broker users retain the documented disconnect safeguard.
- Package discovery does not expose secrets or runtime state.

Explicit deferral

PR 1C proves import and distribution *structure*, not a standalone application installation. The pre-existing dependencies = [] metadata, separation of runtime/dev/test/ML/operations extras, lockfiles with hashes, and a true clean wheel install remain PR 9 scope. The current supported setup continues to install tracked requirements before the project package.

Operational gate

This prerequisite changes no runtime wiring, broker connection, launcher, database, safety state, or order authority. It cannot authorize startup. Gate A remains closed and the trader remains stopped.

PR 2 - Implement a reduce-only safety plane

Staging and current status

PR 2 is deliberately split into two separately reviewed changes:

- **PR 2A / issue #91:** implement the immutable exact-value models, pure

reduce-only policy, durable append-only journal, idempotency and reservation rules, and package-boundary dormancy tests. PR #97 merged this stage as `d17b0d5b4f31ab15e2a9b138cca006c0103b7276`.

- **PR 2B:** integrate the reviewed safety core into active paper execution,

stop-loss, kill-switch, circuit-breaker, and reconciliation paths. This stage has not started and must not be combined with PR 2A review.

PR 2A grants no broker connection or submission authority by itself. The production runner, executor, stop monitor, launcher, and dashboard do not import it. Gate A remains closed, the trader remains stopped, and PR 2B plus the remaining Gate A work are required before any supervised paper start.

Objective

Ensure kill switches and circuit breakers stop new exposure without blocking exits that strictly reduce existing exposure.

Problems addressed

- Kill-switch lock blocks stop-loss orders.
- All-order blocking conflates entry safety with emergency liquidation.
- Emergency shutdown cancels local stops without cancelling broker orders or flattening.
- Stop callbacks and state changes can diverge after failure.

Step-by-step work

1. Define an `OrderIntent` contract containing account, portfolio, symbol, side, quantity, current signed position, target signed position, reason, strategy, reduce-only flag, and idempotency key.
2. Implement a pure exposure-delta validator:
 - long SELL quantity cannot exceed the existing long position;
 - short BUY-to-cover cannot exceed the existing short position;
 - a reduce-only order cannot cross through zero;
 - an absent or uncertain position rejects the order.
3. Split gates into risk-increasing and risk-reducing policies.

4. Make kill switch, daily-loss limit, max-position limit, entry circuit breaker, and closed entry window reject only risk-increasing intents.
5. Keep broker disconnect, uncertain broker position, stale account state, and failed reconciliation as hard blocks even for automated exits; escalate these to an operator rather than guessing.
6. Route stop-loss and future flatten operations through the reduce-only validator.
7. Persist the reason and authorization decision for every rejected and accepted safety order.
8. Make emergency behavior explicit:
 - cancel open entry orders;
 - preserve or replace protective exit orders;
 - never claim flattening unless broker fills prove it.
9. Correct execution failure handling, including the `.message` versus `.msg` mismatch, so rejection details are never hidden by secondary exceptions.

Required tests

- Long reduction, full close, over-close, and reversal attempts.
- Short cover, full cover, over-cover, and reversal attempts.
- Stop execution while kill switch is active.
- Stop execution while daily limits or entry circuit breakers are active.
- Unknown broker position and stale reconciliation cases.
- Repeated identical safety intents proving idempotency.

Done means

- A gap-triggered kill switch cannot block a valid reduce-only exit.
- No reduce-only path can create or reverse exposure.
- Every decision has a durable reason and test coverage.

PR 3 - Establish the canonical market-data contract

Objective

Make every trading, stop, database, strategy, and dashboard consumer use validated, timezone-aware, session-correct market data.

Problems addressed

- RangeIndex values stored as timestamps.
- Database history overwritten on every refresh.
- Historical-bar strings not normalized.
- Regular-hours data used during extended-hours trading.
- Active runner bypasses authoritative validation.
- Stop monitor depends on periodically stale historical closes.
- Concurrent performance timers overwrite one another.

Step-by-step work

1. Define a versioned bar schema: symbol, exchange, timezone-aware timestamp, open, high, low, close, volume, session, source, retrieval time, adjustment state, and quality flags.
2. Parse IBKR dates at the subprocess boundary and reject ambiguous or timezone-naive values.
3. Set the normalized timestamp as the DataFrame index before returning data.
4. Validate OHLC ordering, finite values, nonnegative volume, monotonic order, duplicates, gaps, staleness, and session membership.
5. Store actual timestamps and use explicit upsert rules that preserve history.

6. Make regular versus extended-hours retrieval an explicit strategy/runtime choice.
7. Add an independent quote or real-time bar channel for protective monitoring.
8. Define degraded behavior for stale or missing data:
 - block new entries;
 - retain broker-native protection;
 - alert the operator;
 - never silently substitute the last regular-session close for an extended-hours decision.
9. Make performance timer identifiers unique per symbol and operation instance.
10. Add data lineage and freshness to dashboard/API responses.

Required tests

- DST transitions, market holidays, regular/extended session boundaries, and timezone conversions.
- Duplicate, reversed, NaN, infinite, zero-volume, and out-of-order bars.
- Database accumulation across repeated refreshes.
- Extended-hours fetch and decision tests.
- Stale protective feed behavior.
- Multi-symbol concurrent timer tests.

Done means

- Database timestamps match broker timestamps.
- Historical refreshes accumulate rather than replace numbered rows.
- No strategy or stop consumes unvalidated bars.
- Extended-hours state is explicit and tested.

PR 4 - Make financial state durable and database operations safe

Objective

Create a recoverable financial ledger with constrained schema, safe pooling, safe migrations, and verified backups.

Problems addressed

- Connection-pool replacement can deadlock and requeue closed connections.
- Financial tables lack foreign keys and value constraints.
- SQLite REAL is used for critical money values.
- Trades lack broker order/execution IDs and lifecycle fields.
- Formal multiuser migration is not wired; partial startup migrations swallow failures.
- Migration backup is not WAL-safe and may collapse account rows.
- Deployment and dashboard database settings disagree.
- No demonstrated restore-ready backup exists.

Step-by-step work

1. Select one authoritative database setting and require every runner, dashboard, utility, container, backup, and migration to use it.
2. Fix pool replacement so exactly one valid connection returns to the bounded queue and the internal pool inventory remains correct.
3. Add a schema-version table and explicit ordered migrations. Remove broad exception swallowing.
4. Add database constraints for:
 - valid sides and lifecycle states;

- positive quantities and prices where required;
 - finite, bounded risk percentages;
 - portfolio references;
 - unique idempotency and broker execution identifiers.
5. Enable foreign-key enforcement for every connection.
 6. Store monetary values in lossless integer minor units or validated decimal strings according to a documented convention.
 7. Add append-only tables for order intent, broker order state, fills, commissions, reconciliation snapshots, safety events, and administrator actions.
 8. Make position/account projections derived from confirmed fill events or updated transactionally with those events.
 9. Use SQLite's online backup API or a verified exclusive checkpoint procedure.
 10. Encrypt and rotate off-host backups without exposing broker credentials.
 11. Add automated backup integrity checks and a clean-room restore test.
 12. Create migration dry-run, row-count, checksum, and rollback reports.

Required tests

- Pool failure injection under concurrency.
- Constraint and foreign-key rejection tests.
- Duplicate broker callback/idempotency tests.
- WAL-active backup followed by clean restore.
- Interrupted migration at each step.
- Multiportfolio preservation tests.
- Deployment database path and volume persistence tests.

Done means

- A database exception cannot deadlock the runner.
- An invalid financial row cannot enter through normal application paths.
- Backups include WAL state and restore to an integrity-clean database.
- Every order and fill has durable correlation identifiers.

PR 5 - Add read-only broker reconciliation

Objective

Make broker truth visible and authoritative before any broker-writing capability exists.

Problems addressed

- Startup trusts only the local database.
- Database-load errors fail open to empty positions.
- Broker positions helper exists but is unused by the active runner.
- Open orders, completed orders, executions, cash, and commissions are not reconciled.
- Existing reconciliation documentation points to a missing script.

Step-by-step work

1. Extend the read-only broker adapter to fetch account identity/type, cash, buying power, signed positions, open orders, completed orders, executions, and commissions.
2. Normalize broker objects into versioned domain records.
3. Implement reconciliation at:

- startup;
 - reconnect;
 - fixed periodic intervals;
 - before arming future live capability;
 - after any ambiguous order state in later phases.
4. Compare broker state with the append-only ledger and derived projections.
 5. Classify differences as expected timing lag, recoverable missing event, duplicate event, account mismatch, quantity mismatch, cash mismatch, or unknown.
 6. Quarantine trading on all unknown or material mismatches.
 7. Provide an operator diff that never auto-deletes data.
 8. Persist reconciliation snapshots and resolution actions.
 9. Expose reconciliation age and status on the dashboard.

Required tests

- Clean startup reconciliation.
- Local missing fill recovered from broker execution.
- Duplicate callback.
- Wrong account, wrong quantity, stale cash, unknown order, and reconnect scenarios.
- Database unavailable or partially migrated.
- Operator resolution without destructive replacement.

Done means

- Startup cannot continue trading after local-state uncertainty.
- Every mismatch produces a durable, understandable diff.
- No reconciliation operation deletes history.

Phase B - Repair strategy evidence and execution policy

PR 6 - Rewrite backtesting and walk-forward validation

Objective

Create a deterministic, auditable backtest and replay system that can support strategy decisions without look-ahead or accounting distortion.

Problems addressed

- Returns are always calculated as zero.
- Signals execute at the same bar used to generate them.
- Commissions and spread are double-counted.
- Slippage is unseeded.
- Partial fills are not implemented.
- Exceptions are swallowed and final liquidation is incomplete.
- Metrics assume an inappropriate frequency.
- Walk-forward selection contaminates out-of-sample evidence.
- Critical backtesting modules have no direct tests.

Step-by-step work

1. Reset all engine state at the start of every run and validate sorted, nonempty input.
2. Separate observation time, decision time, submission time, and executable fill time.
3. Default to next-bar or event-realizable fills.
4. Define one cost model where spread, slippage, fees, commissions, borrowing, and market impact are counted exactly once.
5. Seed all stochastic behavior and record the seed in results.
6. Implement liquidity limits and genuine partial fills.
7. Use intrabar OHLC semantics for stop/take-profit triggers while documenting ambiguity resolution.
8. Support long and short accounting, dividends, splits, borrow costs, and delistings as needed by candidate strategies.
9. Calculate equity after fills and append final liquidation/mark-to-market economics.
10. Annualize metrics from actual sampling frequency and guard every zero denominator.
11. Fail the run on data/strategy exceptions unless an explicit, reported policy says otherwise.
12. Implement rolling or nested walk-forward validation with an untouched final holdout.
13. Produce a reproducibility manifest containing code SHA, data checksum, configuration, costs, seed, and model identifiers.

Required tests

- Hand-calculated golden portfolios.
- Look-ahead detection fixtures.
- Commission/spread charged once.
- Partial fill and volume constraints.
- Stop gaps and intrabar ambiguity.
- Empty input, all-error input, and final liquidation.
- Deterministic reruns and untouched holdout verification.

Done means

- Golden fixtures match exact hand calculations.
- Repeated runs with the same manifest are identical.
- No strategy receives launch approval from the old engine.

PR 7 - Unify risk, sizing, exits, and strategy contracts

Objective

Replace duplicate, drifting risk and strategy pathways with one authoritative decision-to-order contract.

Problems addressed

- Hardcoded 10% advanced position cap versus configured 2% cap.
- Symbol validation mistakenly uses the sector limit.
- Daily notional resets after restart.
- Nearest-share rounding can exceed limits.
- Missing market data fails some checks open.
- Existing/short positions are missing or wrong in advanced-risk state.
- Take-profit exists as metadata but not an authoritative lifecycle.
- Pairs legs are non-atomic and use synthetic model training/placeholders.
- Extracted runner modules, `core/engine.py`, and active runner duplicate behavior.

Step-by-step work

1. Select one active strategy interface and one active risk engine.
2. Define a versioned `Signal` -> `OrderIntent` contract containing confidence, expected horizon, source data version, sizing request, entry policy, stop policy, take-profit policy, and expiry.
3. Centralize all configuration loading and remove hardcoded risk values.
4. Enforce position, portfolio, sector, correlation, leverage, liquidity, buying-power, daily notional, churn, and duplicate limits in one place.
5. Floor quantities so calculated exposure never exceeds a cap.
6. Restore daily counters from the ledger on startup.
7. Synchronize long, short, and existing positions from reconciled state.
8. Define take-profit semantics as either broker bracket behavior or an explicitly tested exit policy.
9. Disable incomplete strategies by default, especially pairs, shorts, smart execution, AI discovery, and synthetic-trained selectors.
10. For future pairs support, validate both legs before submission and add combo/hedge, timeout, and compensating-unwind behavior.
11. Archive or clearly quarantine dead alternate engines and unused extracted runner modules until intentionally integrated.
12. Produce a strategy readiness card for every candidate strategy: data, parameters, risk rules, tests, backtest manifest, shadow results, and enabled environments.

Required tests

- Every risk limit across every enabled strategy.
- Restart persistence of daily counters.
- Long and short position accounting.
- Quantity flooring at boundary values.
- Missing data and missing sector/correlation inputs fail closed.
- Two-leg failures and compensating behavior before pairs can be enabled.

Done means

- A configured 2% position cap cannot produce or validate a larger intent.
- Only strategies with readiness cards can be enabled.
- One code path owns sizing, stops, take-profit, and risk validation.

Phase C - Secure remote access and make delivery trustworthy

PR 8 - Build the identity, authorization, and WebSocket security boundary

Objective

Permit safe read-only remote use without exposing broker credentials or granting every user administrative control.

Problems addressed

- Dashboard authentication is disabled locally and uses one shared credential when enabled.
- No active RBAC, MFA, portfolio ownership, or individual actor audit.
- Kill-switch reset/start/stop use ordinary dashboard credentials.
- Shared WebSocket token allows consumer-to-producer impersonation.
- Token is placed in HTML/query strings; null/missing origins are admitted.
- Basic auth can be exposed by direct published ports.
- Legacy unsalted SHA-256 password setup.
- Model signature enforcement uses inconsistent mode detection.

Step-by-step work

1. Introduce individual identities with strong password hashing and MFA.
2. Define roles such as viewer, portfolio operator, safety operator, and administrator.
3. Enforce portfolio-level authorization on every HTTP and WebSocket response.
4. Use secure server-side sessions or scoped, short-lived tokens with rotation and revocation.
5. Require reauthentication, a reason, and elevated permission for start, stop, kill-switch reset, and future live arming.
6. Consider two-person approval for live kill-switch reset.
7. Write administrative actions to append-only audit storage before applying the change where safety permits.
8. Replace shared producer trust with local IPC, mTLS, or a separate producer credential unavailable to dashboard consumers.
9. Use same-origin WSS through the TLS proxy; remove query-string tokens and reject null/missing origins for remote deployment.
10. Add message schemas, size limits, rate limits, subscription authorization, and portfolio filtering.
11. Move all secrets to approved environment/secret-manager injection and implement rotation procedures.
12. Make model signing mandatory from the canonical runtime contract and deserialize the exact verified bytes.
13. Add recursive log/config redaction and stop logging broker account identifiers except approved aliases.

Required tests

- Role and portfolio access matrix.
- Session revocation, expiration, MFA, and reauthentication.
- CSRF, CORS, origin, proxy, brute-force, and rate-limit tests.
- Consumer attempts producer impersonation.
- Query-token leakage and log-redaction tests.
- Unsigned/tampered model refusal in every production-like mode.

Done means

- Remote users are individually attributable and least-privileged.
- Dashboard consumers cannot publish runner events.
- Safety actions have elevated controls and durable audit.
- Mobile access may begin only as read-only.

PR 9 - Restore CI, packaging, dependency, and supply-chain truth

Objective

Make a green build mean the installable artifact, critical tests, security checks, and supported runtime all passed.

Problems addressed

- 551 current mypy errors and formatting/lint failures.
- False-green tests return booleans instead of asserting.
- Safety tests are ignored by default.
- Critical modules have zero coverage; `app.py` is excluded.
- Performance jobs are no-ops.
- Package metadata omits runtime dependencies.
- Security scans do not always scan installed application dependencies.
- CI actions and container images use mutable references.
- Model/runtime dependency versions drift.

Step-by-step work

1. Choose one authoritative CI workflow and remove contradictory duplicates.
2. Preserve PR 1C's precise subpackage discovery and package-data allow-list.
3. Declare runtime dependencies and split dev, test, ML, and operations extras.
4. Create a reproducible lock with hashes for supported Python versions.
5. Build a wheel and test it in a clean environment rather than relying on checkout imports.
6. Convert return/print-based tests to assertions and remove unconditional success messages.
7. Stop ignoring the safety suite; rewrite it into deterministic tests.
8. Add direct tests for backtester, order lifecycle, reconciliation, stop safety, database failures, and deployment startup.
9. Include dashboard/API code in coverage and set risk-based coverage thresholds.
10. Resolve type errors in active safety paths first; maintain a shrinking, explicit debt budget for lower-risk legacy code.
11. Add real performance, leak, concurrency, and soak jobs.
12. Scan the resolved production dependency set and built image.
13. Pin third-party actions and images to reviewed SHAs/digests.
14. Generate SBOM and provenance, sign artifacts, and verify them during deployment.
15. Pin model-training and inference environments and reject incompatible serialized artifacts.

Required tests

- Clean wheel install and smoke test.
- Exact locked environment test matrix.
- Coverage gates on critical modules.
- Dependency and container vulnerability policy.
- Reproducible artifact/hash comparison.
- No pending async tasks or leaked processes at test completion.

Done means

- Required CI is green with no placeholder jobs.
- A clean wheel contains every required module and dependency.
- Critical safety paths have meaningful assertion coverage.

PR 10 - Make telemetry, alerts, and the dashboard authoritative

Objective

Give operators a truthful, fresh, source-attributed view of the actual runner and broker state.

Problems addressed

- Dashboard panels contain hardcoded/mock metrics.
- Missing runner data is presented as healthy zero/default state.
- Dashboard process-local objects do not reflect runner state.
- Mode/status can be stale or contradictory.
- Webhook alerts are disabled in the current environment.
- Logs are tracked in Git and may contain sensitive trading telemetry.
- Mobile layout, accessibility, and polling behavior are weak.

Step-by-step work

1. Define versioned presentation contracts for runtime, broker, orders, fills, positions, risk, strategies, data freshness, reconciliation, protection, alerts, and build identity.
2. Include source, timestamp, age, portfolio, environment, and availability state in every response.
3. Persist runner telemetry to a shared durable store or event stream rather than accessing process-local objects.
4. Remove every hardcoded or estimated value presented as actual execution/risk state.
5. Render `Unavailable`, `Stale`, `Degraded`, `Mock`, and `Zero` distinctly.
6. Add non-dismissible mode/account/reconciliation/protection banners.
7. Expose last completed cycle, last successful broker query, last fill, data age, current kill-switch reason, and active broker protective orders.
8. Test email/SMS/pager/webhook alerts end-to-end and add a dead-man alert.
9. Add tamper-evident audit event access for authorized operators.
10. Remove tracked runtime logs from future commits and document safe history-cleaning as a separate user-approved operation if desired.
11. Use same-origin WSS, responsive breakpoints, accessible tabs, and active-tab/subscription-driven refresh.
12. Keep the initial mobile surface read-only.

Required tests

- API contract and freshness tests.
- Runner stopped, stale, DB error, broker disconnected, alert failed, and reconciliation mismatch states.
- Browser tests for desktop/mobile layouts and keyboard navigation.
- Verified human receipt of every critical alert class.

Done means

- No mock or unavailable value is displayed as real/healthy.
- Operators can determine mode, account, freshness, reconciliation, and protection at a glance.
- Critical alerts reach a human and are recorded.

Phase D - Add live capability behind a disabled gate

PR 11 - Implement one live IBKR order lifecycle adapter

Objective

Create a complete broker order state machine while keeping the capability disabled in all normal environments.

Problems addressed

- Dormant `LiveExecutor` schedules orders ambiguously and treats `Submitted` as filled.
- No active broker `place_order` implementation.
- No idempotent order reference or authoritative broker lifecycle.
- No partial fill, reject, cancel, replace, commission, reconnect, or unknown-state handling.
- Order state is updated from simulation rather than broker fills.

Step-by-step work

1. Define one asynchronous broker interface. Remove or quarantine synchronous wrappers that can outlive returned results.
2. Use deterministic client order IDs/order references tied to durable order intents.
3. Model states: created, pending submit, pre-submitted, submitted, partially filled, filled, cancel pending, cancelled, rejected, inactive, unknown, and reconciliation required.

4. Persist broker order ID, permanent ID, execution ID, timestamps, quantities, prices, commissions, warnings, and rejection details.
5. Update positions, cash, risk, and dashboard only from broker-confirmed fills.
6. Make retries idempotent and ambiguity-safe. A timeout must transition to unknown/reconcile, never automatic duplicate submission.
7. Add account allowlist and contract qualification.
8. Enforce market-hours, order-type, notional, buying power, shortability, and exchange/session constraints.
9. Implement cancel and replace with reconciliation after reconnect.
10. Cancel open entry orders during safety shutdown while preserving reduce-only protection.
11. Keep the entire adapter behind a disabled capability gate and use an IBKR test/paper account for integration tests.

Required tests

- Submit acknowledgement without fill.
- Partial fills and multiple callbacks.
- Reject, inactive, cancel, replace, timeout, reconnect, duplicate callbacks, and duplicate client IDs.
- Wrong account and contract ambiguity.
- Process crash between broker fill and local persistence.
- No position change before confirmed fill.

Done means

- Submitted is never treated as a fill.
- Ambiguous outcomes cannot produce automatic duplicate orders.
- Broker fills are the only source of financial state transitions.
- Capability remains disabled outside explicit integration tests.

PR 12 - Add broker-native protective orders and exit lifecycle

Objective

Ensure every live position has broker-resident protection that survives local failures.

Problems addressed

- In-memory synthetic stops disappear when process/host/Gateway fails.
- Stop prices can be stale and gap behavior is unsafe.
- Take-profit is not an authoritative execution feature.
- Emergency flattening and broker cancellation are incomplete.

Step-by-step work

1. Define supported bracket/OCO structures for long and short positions.
2. Implement IBKR parent/child transmit sequencing so protection cannot be accidentally left unsubmitted.
3. Establish stop-market versus stop-limit policy; default gap protection must not create an unfillable stale limit.
4. Implement broker-native take-profit where strategy policy calls for it.
5. Reconcile protective orders on startup, reconnect, fill, partial fill, quantity change, and cancel/replace.
6. Implement trailing-stop modification with rate limits and broker acknowledgement.
7. Define overnight, extended-hours, outside-RTH, and corporate-action behavior.
8. Prevent entry completion from being considered safe until expected protection is acknowledged at the broker.
9. Add operator-visible protection status and emergency cancel/flatten runbooks.
10. Retain synthetic monitoring only as an independent alarm, not the primary protection.

Required tests

- Parent fill before/after child acknowledgements.
- Partial fills and protective quantity updates.
- Gap through stop, reconnect, Gateway failure, process kill, and host restart.
- Manual broker-side cancellation detection and repair.
- OCO behavior and trailing modifications.

Done means

- Broker UI proves protection remains active with the RoboTrader processes stopped.
- Every reconciled live position has the required protection or trading is quarantined.

Phase E - Prove operations, then allow only a limited launch

PR 13 - Failure injection, restoration, and multi-week paper soak

Objective

Prove that the complete system behaves safely under realistic failures before enabling live capability.

Step-by-step work

1. Build a broker simulator covering submit, acknowledge, partial fill, fill, reject, cancel, duplicate callback, disconnect, and reconnect.
2. Run controlled failures for:
 - stale/malformed market data;
 - Gateway restart and 2FA delay;
 - network partition;
 - process kill and host reboot;
 - database locked/corrupt/full;
 - disk full and log growth;
 - alert provider failure;
 - duplicate executions;
 - missing or cancelled protective orders;
 - second-leg pairs failure if pairs remains planned.
3. Rehearse online backup and clean-machine restore.
4. Rehearse kill switch, cancel open entries, and manual flatten using the paper broker.
5. Verify reconciliation after every failure.
6. Conduct a multi-week paper/shadow soak using the exact release artifact and operational topology.
7. Track restarts, stale data, reconciliation mismatches, duplicate order attempts, pending tasks, DB latency, alert latency, and unbounded logs.
8. Produce a signed launch-readiness evidence package.

Done means

- Zero unexplained duplicate orders.
- Zero unresolved reconciliation drift.
- Every active paper position has expected protection behavior.
- Restore and alert drills succeed.
- Soak exits within agreed reliability and risk thresholds.

PR 14 - Deliver the selected remote/mobile/cloud topology

Objective

Support remote or cloud operation only after an explicit architecture decision and without creating multiple order writers.

Step-by-step work

1. Decide whether macOS + IBC remains the only order-writing topology.
2. If cloud is not required, remove misleading Kubernetes/live deployment artifacts and publish a secure remote read-only dashboard design.
3. If cloud is required, design:
 - IB Gateway ownership and interactive 2FA;
 - single active writer with lease and fencing;
 - encrypted persistent database and backups;
 - secret-manager integration;
 - private network and NetworkPolicy;
 - immutable signed images;
 - real liveness/readiness endpoints;
 - rolling deploy prevention for the order writer;
 - rollback and disaster recovery.
4. Make Docker paper mode boot and persist correctly before any production manifest.
5. Replace placeholder deployment jobs with real staging, smoke, health, approval, production, and rollback operations.
6. Release mobile as read-only first. Add remote mutating actions only after separate threat modeling and approval.

Done means

- Exactly one fenced order writer can exist.
- Health checks measure actual runner/broker/reconciliation readiness.
- Deployment and rollback have been exercised, not merely documented.
- Remote clients use strong identity and TLS without direct broker credential access.

PR 15 - Limited live canary and staged expansion

Objective

Enable the smallest reasonable real-money exposure only after all prior gates pass.

Preconditions

- PR 1, PR 1A, PR 1B, and PRs 2 through 14 are complete as applicable.
- PR 1B reconciliation evidence was reviewed and did not itself mutate or authorize mutation of the ledger or safety state.
- All P0 and P1 audit findings are closed.
- Independent security and trading-safety review approves the evidence package.
- Reconciliation is clean.
- Broker-native protection is visible in IBKR.
- Human alerts, backup restore, and manual flatten drills pass.
- Multi-week paper/shadow soak passes.

Step-by-step work

1. Create separate live account configuration, credentials, database, logs, model artifacts, and deployment identity.
2. Require a manual arming ceremony with named operator, reason, build SHA, configuration fingerprint, account confirmation, and expiry time.
3. Start with:
 - tiny symbol allowlist;
 - one open position maximum;
 - very low absolute notional and daily-loss caps;
 - long-only simple orders;
 - no pairs, shorts, smart execution, AI discovery, extended-hours entries, or automatic strategy expansion.
4. Require broker-native stop protection before the entry is considered operationally complete.
5. Monitor every order and fill in real time with human acknowledgement.
6. Automatically disable new entries on any reconciliation, data, alert, protection, or process-health degradation.
7. Review after each trade and each day. Expansion requires a new approved stage, never an automatic threshold change.
8. Maintain a tested manual cancel/flatten path and a documented return-to-disabled procedure.

Done means

- Canary trades reconcile exactly with broker records.
- No safety or operational deviation is unexplained.
- Expansion is separately approved with evidence.

6. Launch gates

Launch gates are cumulative. Gate B and every later gate require Gate A, including explicit PR 1A correlation evidence and PR 1B read-only reconciliation evidence. A numeric PR range never implicitly omits PR 1A or PR 1B.

Gate A - Supervised local paper readiness

Required PRs: 1, 1A, 1B, 2 through 5, plus relevant parts of 7.

Evidence required:

- Broker confirms paper account and read-only API.
- PR 1A failure injection proves delayed, mismatched, stale, or uncorrelated

broker data cannot reach valuation, risk, strategy, persistence, or stops.

- PR 1B broker-versus-ledger reconciliation is current, reviewed, and proves it

did not modify the database or safety state.

- Paper state cannot collide with any future live state.
- Reduce-only exits pass all blocking-state tests.
- Data timestamps, freshness, and session semantics are correct.
- Startup fails closed on database or reconciliation uncertainty.
- Backups restore cleanly.
- No unsafeguarded destructive utility remains.

Gate B - Strategy evaluation readiness

Required PRs: 6 and 7.

Evidence required:

- Golden backtest accounting passes.
- No same-bar look-ahead.
- Costs are counted once.
- Results reproduce from a manifest.
- Candidate strategy has a readiness card and untouched holdout results.
- Risk and sizing caps are proven across all paths.

Gate C - Remote/mobile read-only readiness

Required PRs: 8 through 10.

Evidence required:

- Identity, MFA, least privilege, portfolio authorization, TLS, and revocation tests pass.
- WebSocket consumers cannot impersonate the producer.
- Mobile is read-only.
- Dashboard never presents stale/mock/unavailable values as healthy facts.
- Alerts reach a human.

Gate D - Live implementation readiness

Required PRs: 11 and 12.

Evidence required:

- Complete order lifecycle works in IBKR paper integration.
- Submitted is distinct from filled.
- Ambiguous timeouts reconcile without duplicate submission.
- Broker-native protection survives process and host failure.
- Broker truth drives all financial state.

Gate E - Live canary readiness

Required PRs: Gate A through Gate D, then PRs 13 through 15.

Evidence required:

- Failure drills, backup restore, alerts, and multi-week soak pass.
- Supported deployment topology is explicit and tested.
- Independent safety/security approval is recorded.
- Canary constraints and manual arming are active.

7. Cross-cutting problem register

Use these identifiers in issues and PR descriptions.

- RT-001: Active runtime always uses PaperExecutor; no coherent live path.
- RT-002: Dormant LiveExecutor has ambiguous async behavior and treats Submitted as success.
- RT-003: No broker-authoritative startup/reconnect reconciliation.
- RT-004: Kill-switch lock blocks reduce-only stop exits.
- RT-005: Stops are synthetic and periodically stale.
- RT-006: No broker-native bracket/OCO protection or authoritative take-profit.
- RT-007: Position limits drift between configuration and risk implementations.

- RT-008: Daily counters and advanced-risk state do not restore correctly.
- RT-009: Pairs execution is non-atomic and selector logic contains synthetic placeholders.
- RT-010: Market-data timestamps are stored as RangeIndex values.
- RT-011: Extended-hours decisions may consume regular-hours-only data.
- RT-012: Active data validation and performance telemetry are incomplete.
- RT-013: Backtest returns, costs, execution timing, and walk-forward evidence are invalid.
- RT-014: Destructive utilities can erase databases or positions.
- RT-015: Database pool error path can deadlock.
- RT-016: Financial schema lacks broker identifiers, strong constraints, and lossless values.
- RT-017: Migration and backup behavior is not WAL-safe.
- RT-018: Database paths differ across runner, dashboard, Compose, Kubernetes, and backup tools.
- RT-019: No verified automated off-host backup and restore program.
- RT-020: Shared dashboard credential lacks authorization and individual attribution.
- RT-021: Kill-switch reset/start/stop lack elevated approval and durable audit.
- RT-022: Shared WebSocket token allows producer impersonation.
- RT-023: Model signing and runtime-version enforcement can fail open.
- RT-024: Logging and Git history expose sensitive trading telemetry.
- RT-025: Tests have false-green patterns, ignored suites, and critical coverage gaps.
- RT-026: CI, packaging, dependency scanning, and supply-chain controls are not release-grade.
- RT-027: Docker/Kubernetes paths bypass preflight and do not persist the actual ledger.
- RT-028: Deployment workflows contain placeholder deploy, smoke, and rollback steps.
- RT-029: Dashboard contains hardcoded/mock or process-local operational values.
- RT-030: Mobile/cloud transport, accessibility, and topology are incomplete.
- RT-031: Duplicate unfinished engines and runner modules create architecture drift.
- RT-032: Documentation claims conflict with executable readiness.
- RT-033: Uncorrelated or timed-out broker responses can be assigned to the

wrong symbol and reach risk, stop, valuation, or persistence consumers.

8. Progress register

Update after each merge.

- Phase 0 CI truth gate: PR #83 merged on 2026-07-23 (7f5de0a)
- Phase 0 runtime-stability prerequisite: PR #81 merged on 2026-07-23 (b7e5005)
- PR 1: PR #82 merged on 2026-07-23 as 393f533. Local evidence: 1,046

passed, 5 skipped, 42% total coverage; Black, isort, Flake8, Bandit, pip integrity, shell syntax, YAML parsing, and diff checks passed. Hosted CI passed Python 3.10 through 3.12 tests, production unit/integration/performance matrices, lint, code quality, security, Trivy, Docker build, container structure, and Docker Compose containment. All 15 review threads were resolved. A two-phase review examined 11 initial findings: nine were confirmed and remediated, one dashboard Lsof diagnostic was downgraded and remediated, and RT_STATE_NAMESPACE file-path isolation was safely deferred because changing the legacy paper kill-switch path could bypass the currently triggered state. The challenger then rejected three successive lifecycle designs until startup ordering, the operator-facing Gateway CLI, concurrent launcher/recovery races, and lock ownership were all fail-closed. The final design acquires one kernel advisory lock, transfers it to the launcher with an inherited descriptor, validates that descriptor before runtime work, and prevents Gateway, dashboard, or runner descendants from retaining it. The final independent review passed. The active runner already rejects backtest mode; separate non-paper safety state remains required before that mode may use shared risk components. The external Claude review action did not review code because its configured credential returned HTTP 401 with zero tokens; this infrastructure failure is recorded on PR #82 and was not treated as repository validation. PR #80 (dd26ad5, edd0288) is explicitly superseded: no commit from that branch

was merged. Its 11 review threads are mapped one-to-one to PR 1 / PR #82, PR 6, or PR 11 in the branch disposition record. Separate branch requirements, rather than review threads, are retained for PRs 8 and 10.

- PR 1A: PR #87 merged on 2026-07-23 as

4caf7b782cbf43ff4397f1b89b42d5f657ecee8e from exact reviewed head aa62b3e20dbd88096aa78a5875a8dc48e298f7ee. Local focused validation passed 371 tests with 2 skipped; the local full suite passed 1,396 tests with 5 skipped and 20 warnings. All repository-owned hosted checks were green, Codex exact-head review was clean, and the independent challenger returned PASS. The external Claude run 30047548445 provided no validation: its revoked OAuth credential failed with HTTP 401 after using zero tokens and incurring zero cost. PR 1A closes the incident-driven broker-correlation, event-time, transport-poisoning, stop-protection, and fail-closed lifecycle scope. It does not authorize a restart. Gate A remains closed, the trader remains stopped, and PR 1B read-only reconciliation is next.

- PR 1B: PR #90 merged on 2026-07-23 as

0d43006561071e27b217cb6d16f3c0a245a18655 from exact reviewed head ae133c054721ea8ca656594053594e0ae43649d1. The local full suite passed 1,564 tests with 5 skipped and 20 warnings. The final strict whole-PR review returned PASS after 314 focused tests; the client-ID boundary review returned PASS after 151 focused tests. All repository-owned hosted checks passed, including Python 3.10 through 3.12 tests, production unit/integration/performance matrices, lint, security, Docker, import validation, Trivy, and SARIF upload. Earlier Codex reviews found cleanup and shared client-ID compatibility defects; both were fixed and all threads were resolved. The final-head Codex request could not run because the account reached its code-review usage limit. The external Claude action again provided no validation because its revoked OAuth credential returned HTTP 401 with zero tokens and zero cost.

The first real command was run from merged code against the stopped local paper runtime. It did not connect to the broker: runtime validation blocked because `.env` lacks `IBKR_ACCOUNT`, `IBKR_APPROVED_ACCOUNTS`, and `IBKR_ACCOUNT_TYPE`. The report stated `mutated_state=false` and `authorizes_startup=false`; independent before/after hashes of `.env`, the ledger and SQLite sidecars, kill-switch state and lock, bypass log, and trading log were unchanged. Issue #92 requires the operator to configure the exact paper account and dedicated reconciliation client ID locally without publishing the raw account number. Reconciliation remains incomplete, no data or safety state was corrected, and this merge does not authorize startup.

- PR 1C: PR #95 merged on 2026-07-24 as

dff4c8b597a54c614d8925565f28aa865f8ae676 from exact reviewed head 431513b9c7034ac2712ffb54acb58429e04281ba. The focused package, import-isolation, and broker-boundary suite passed 76 tests. The full local suite passed 1,568 tests with 5 skipped and 20 known warnings. All repository-owned hosted checks passed, including Python 3.10 through 3.12, production unit/integration/performance matrices, lint, security, containers, build, BugBot, and Trivy. The prior missing-template review finding was fixed, regression-tested, and its only review thread was resolved.

Exact-head two-phase review ran code-quality, bug, trading-safety, and style passes plus a verification challenger. Reviewers reproduced that the pre-existing empty runtime dependency metadata prevents a standalone clean wheel install; the challenger correctly retained it as medium PR 9 debt rather than copying the current mixed requirements into this prerequisite. The challenger also downgraded the offline no-build-isolation backend concern to optional low-priority test hardening because normal PEP 517 builds honor the `setuptools>=83.0.0` floor and supported CI/dev setup pins 83.0.0. Test-only Bandit assert and shell-free subprocess notices were filtered as false positives. Final two-phase verdict: PASS with no blocking finding.

The external Claude action did not review code: its credential returned HTTP 401 before inference with zero input tokens, zero output tokens, and zero cost. Final-head Codex and Cursor review requests reported usage limits; those unavailable reviews were recorded rather than counted as passes. Issue #94 closed and the source branch was deleted. PR 1C changes no runtime wiring or order authority. Gate A remains closed, the trader remains stopped, and PR 2A / issue #91 merged through PR #97 as d17b0d5b4f31ab15e2a9b138cca006c0103b7276. The dormant merge grants no startup or order authority.

- PR 2: Staged as dormant PR 2A (issue #91) followed by separately reviewed

runtime-integration PRs 2B.1 through 2B.3. PR #97 merged PR 2A on 2026-07-25 from exact reviewed head 5cf0e4ec48178fdecde6794e920d6af7c67b58f8 as d17b0d5b4f31ab15e2a9b138cca006c0103b7276. PR #104 merged PR 2B.1 on 2026-07-25 from exact reviewed head 0d5585b46f8f1b495d944e24b23df5a7c01cfc2d as 3ecd0a05b3352ddcd4519662b0fe957751f3fdb1. PR 2B.2 is next; PR 2B.3 remains pending.

PR 2A contains strict exact-Decimal models, account/portfolio-aware reduce-only validation, zero-crossing and over-close rejection, deterministic idempotency, one-shot submission permits, dual-scope active reservations, crash/unknown-outcome quarantine, exact terminal reconciliation, and a dedicated append-only SQLite hash-chain journal. Journal initialization is explicit, owner-only, rejects unrelated databases and symlinked final paths, and binds the actual SQLite-owned native descriptor to the independently opened journal device/inode around reads and mutations. The descriptor proof fails closed outside supported GIL-enabled CPython 3.10 through 3.14 with a default Unix SQLite VFS. Tests also prove that importing `robo_trader.safety` does not import or wire any production runtime.

Current PR evidence: the focused safety and package-boundary suite passes 103 tests. The full repository suite passes 1,667 tests with 5 skipped and 20 known warnings. Black, isort, Flake8, and Bandit pass for the new package and tests. Independent code, bug, trading-safety, style, and challenger reviews passed before GitHub review. GitHub Codex then identified three valid gaps: stale plan

status, direct-model zero-crossing acceptance, and existing-symlink journal redirection. All three are remediated on the branch with regression coverage. Post-fix adversarial review then exposed same-schema swap-open-restore races in both read and write paths, a false-attribution weakness in process-wide descriptor enumeration, a callback self-deadlock, and unsafe CPython ABI assumptions. The final design compares the native descriptor owned by SQLite itself with an independent `O_NOFOLLOW` guardian, rejects unsupported interpreter/VFS builds before pointer access, and has focused substitution, decoy-descriptor, ABI-guard, reentrant-callback, and repeated concurrency coverage. A final independent review also found and verified the repair of a post-bind exception cleanup leak; the regression proves the binding map, SQLite connection, and guardian descriptor are all released.

Exact-head hosted CI passed every repository-owned build, Docker, container-structure, Compose, lint, security, Trivy, code-quality, bug-detection, and Python 3.10 through 3.12 unit/integration/performance job. Exact-head Cursor security review found no medium, high, or critical finding, and all three earlier GitHub Codex threads were fixed, replied to, and resolved. A requested exact-head Codex rerun was unavailable because the account reached its code-review usage limit. The external Claude action did not review code because its revoked OAuth token returned HTTP 401 before inference with zero input tokens, zero output tokens, and zero cost. Those unavailable external reviews were recorded rather than counted as passes.

This stage remains dormant and cannot authorize startup or order placement. Gate A remains closed and the trader remains stopped.

PR 2B is split into PR 2B.1 (issue #100: paper runtime identity, trusted evidence, and startup replay), PR 2B.2 (issue #101: route paper exits and separate hard/soft gates), and PR 2B.3 (issue #102: exact settlement and crash/restart quarantine). PR 2B.1 is merged. It adds an identity-bound journal, read-only startup replay before any supervised process/Gateway mutation, one-transaction cross-portfolio allocation evidence, connected-generation qualified-contract lineage, and a typed producer-evidence assembly boundary. The coordinator has only a sealed in-memory fake submitter: no production order path is wired in PR 2B.1. Its broker snapshot wrappers are explicitly dormant integration-test scaffolding, not an operational trust source. PR 2B.2 must replace them with account-bound snapshots emitted by the broker client, bind allocation evidence to the validated runtime ledger identity and opened database device/inode, and revalidate live transport, account positions, and all-client open orders immediately before broker dispatch. These are high-severity prerequisites before the runner may call authorization or any real submitter may consume a permit. The dormancy regression fails if the runner or any active order path imports that assembly module or calls the coordinator's authorization/submission methods.

PR 2B.1 schema v2 intentionally fails closed on a schema-v1 PR 2A journal. No operational v1 journal is known and none exists in the inspected runtime, so this is not a dormant-PR blocker. Any discovered v1 file must be preserved; a later migration must use an explicit backup/copy-and-verify procedure and must never rebind or rewrite the only copy. PRs 2B.2, 2B.3, 3 through 5, and relevant PR 7 work remain outstanding.

Final PR 2B.1 evidence: after two late GitHub Codex findings, the configured journal path preserves its lexical final component so the journal can reject symlink substitution itself, and Python dependencies are bootstrapped before journal verification without importing side-effectful RoboTrader runtime modules. Dedicated tests prove same-identity symlink rejection, fresh and incomplete virtual-environment bootstrap, and fail-closed handling of a partially successful dependency installation. Two independent reviews returned PASS on the repaired exact tree. The focused late-fix suite passed 77 tests; the final full local suite passed 1,801 tests with 5 expected skips and 20 known warnings. Black, isort, Flake8, Bash syntax, and diff checks passed.

All 26 repository-owned exact-head hosted checks passed: Python 3.10 through 3.12, production unit/integration/performance jobs, lint, security, Docker, container structure, Compose, build, Trivy, BugBot, and the Cursor vulnerability scan. Both late Codex review threads were replied to and resolved. The external Claude action provided no validation because its revoked OAuth token returned HTTP 401 before inference with zero input or output tokens. That unavailable review was recorded rather than counted as a pass. The merge grants no order authority or startup approval. Gate A remains closed and the trader remains stopped.

- PR 3: Not started
- PR 4: Not started
- PR 5: Not started
- PR 6: Not started
- PR 7: Not started
- PR 8: Not started
- PR 9: Not started
- PR 10: Not started
- PR 11: Not started
- PR 12: Not started
- PR 13: Not started
- PR 14: Not started
- PR 15: Not started

9. Standard PR evidence checklist

Every safety-relevant PR must include:

- problem identifiers addressed;
- explicit non-goals;
- design or threat model where appropriate;
- migration and rollback behavior;
- unit, integration, and failure-injection tests;
- exact commands and results;
- database backup and restore implications;
- paper/live separation implications;
- security and credential implications;
- operator/dashboard implications;
- updated runbook and documentation;
- before/after screenshots for operator-facing changes;
- reviewer sign-off from trading safety and security/data integrity;
- confirmation that no user trading data was deleted.

10. Recommended verification commands

Commands must be adjusted as the CI contract evolves, but the starting set is:

```
.venv/bin/python -m pytest tests/ -q
.venv/bin/python -m pytest tests/ -q --cov=robo_trader --cov-report=term
.venv/bin/python -m mypy robo_trader
.venv/bin/python -m black --check .
.venv/bin/python -m flake8
.venv/bin/python -m bandit -r robo_trader scripts -ll
.venv/bin/python -m pip check
python3 scripts/preflight_check.py --verbose
python3 scripts/gateway_manager.py status
lsof -nP -iTCP:4002 -sTCP:LISTEN
lsof -nP -iTCP:4002 -sTCP:CLOSE_WAIT
```

Never use a test command that can access the production database unless test isolation has been verified for that exact command.

11. Final launch decision template

The launch approver must answer all of the following with evidence:

1. Which exact build, configuration fingerprint, model artifacts, account, database, and topology are being armed?
2. Is broker reconciliation clean now?
3. Are broker-native protective orders verified?
4. Are all risk limits loaded from the canonical configuration and displayed correctly?
5. Can reduce-only exits execute while entry trading is blocked?
6. Did backup restoration and manual cancel/flatten drills pass?
7. Are human alerts working?
8. Did the required paper/shadow soak pass without unexplained drift?
9. Are identity, authorization, WebSocket, and audit controls active?
10. Is the live canary restricted to the approved account, symbols, position count, notional, and expiry?

If any answer is no, unknown, stale, or based only on documentation rather than observed evidence, live trading remains disabled.